

Answers: Introduction to set theory

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Summary

Answers to questions relating to the guide on introduction to set theory.

These are the answers to [Questions: Introduction to set theory](#).

Please attempt the questions before reading these answers!

Q1

- 1.1. This set is finite.
- 1.2. This set is countably infinite.
- 1.3. This set is finite.
- 1.4. This set is uncountable.
- 1.5. This set is uncountable.
- 1.6. This set is countably infinite.

Q2

- 2.1. This is TRUE.
- 2.2. This is FALSE, as $2 \notin \{1, 3, 5, 6, 7, 16, 18, 20\}$.
- 2.3. This is FALSE, as $\frac{1}{3}, \frac{41}{2} \notin \mathbb{Z}$.
- 2.4. This is TRUE.
- 2.5. This is FALSE, as $(1.6, 5]$ is open on the left, so $1.6 \notin (1.6, 5]$.
- 2.6. This is TRUE.
- 2.7. This is FALSE, as 15 is not even, so $15 \notin \{x : \frac{x}{2} \in \mathbb{Z}\}$.
- 2.8. This is TRUE, as 6, 18, 24 and 27 are all multiples of 3.

Q3

3.1.

$$\begin{aligned} & \{ \emptyset, \\ \mathcal{P}(A) = & \{1\}, \{2\}, \{3\}, \\ & \{1, 2\}, \{1, 3\}, \{2, 3\}, \\ & \{1, 2, 3\} \end{aligned}$$

3.2.

$$\begin{aligned} & \{ \emptyset, \\ \mathcal{P}(B) = & \{\frac{1}{2}\}, \{\frac{1}{3}\}, \{\frac{1}{6}\}, \\ & \{\frac{1}{2}, \frac{1}{3}\}, \{\frac{1}{2}, \frac{1}{6}\}, \{\frac{1}{3}, \frac{1}{6}\}, \\ & \{\frac{1}{2}, \frac{1}{3}, \frac{1}{6}\} \end{aligned}$$

3.3.

$$\begin{aligned} & \{ \emptyset, \\ & \{1\}, \{3\}, \{6\}, \{9\}, \\ \mathcal{P}(C) = & \{1, 3\}, \{1, 6\}, \{1, 9\}, \{3, 6\}, \{3, 9\}, \{6, 9\}, \\ & \{1, 3, 6\}, \{1, 3, 9\}, \{1, 6, 9\}, \{3, 6, 9\}, \\ & \{1, 3, 6, 9\} \end{aligned}$$

3.4.

$$\begin{aligned} & \{ \emptyset, \\ \mathcal{P}(D) = & \{\emptyset\}, \{4\}, \\ & \{\emptyset, 4\} \end{aligned}$$

3.5.

$$\begin{aligned} & \{ \emptyset, \\ & \{4.5\}, \{8\}, \{9\}, \{10\}, \\ \mathcal{P}(E) = & \{4.5, 8\}, \{4.5, 9\}, \{4.5, 10\}, \{8, 9\}, \{8, 10\}, \{9, 10\}, \\ & \{4.5, 8, 9\}, \{4.5, 8, 10\}, \{4.5, 9, 10\}, \{8, 9, 10\}, \\ & \{4.5, 8, 9, 10\} \end{aligned}$$

3.6.

$$\begin{aligned} & \{ \emptyset, \\ & \quad \{\sqrt{2}\}, \{5\}, \{\frac{16}{3}\}, \{18.2\}, \\ \mathcal{P}(F) = & \quad \{\sqrt{2}, 5\}, \{\sqrt{2}, \frac{16}{3}\}, \{\sqrt{2}, 18.2\}, \{5, \frac{16}{3}\}, \{5, 18.2\}, \{\frac{16}{3}, 18.2\}, \\ & \quad \{\sqrt{2}, 5, \frac{16}{3}\}, \{\sqrt{2}, 5, 18.2\}, \{\sqrt{2}, \frac{16}{3}, 18.2\}, \{5, \frac{16}{3}, 18.2\}, \\ & \quad \{\sqrt{2}, 5, \frac{16}{3}, 18.2\} \} \end{aligned}$$

Q4

4.1. $|\mathcal{P}(A)| = 2^{|\mathcal{P}(A)|} = 2^3 = 8$

4.2. $|\mathcal{P}(B)| = 2^{|\mathcal{P}(B)|} = 2^3 = 8$

4.3. $|\mathcal{P}(C)| = 2^{|\mathcal{P}(C)|} = 2^4 = 16$

4.4. $|\mathcal{P}(D)| = 2^{|\mathcal{P}(D)|} = 2^2 = 4$

4.5. $|\mathcal{P}(E)| = 2^{|\mathcal{P}(E)|} = 2^4 = 16$

4.6. $|\mathcal{P}(F)| = 2^{|\mathcal{P}(F)|} = 2^4 = 16$

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v1.0: initial version created 09/24 by tdhc.

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